

SJM UNIVERSITY

SJM INSTITUTE OF TECHNOLOGY

Centre of Excellence Lab • AIML Department

OFFICIAL INVITATION

Dear Students,

You are cordially invited to the Inaugural Function of the **21-Day Hands-on Training Programme on "Cyber Security"**, organized in association with **ICT Academy, Redington Foundation, and CISCO**.

Your presence will add value and encouragement to this important training initiative. This cutting-edge program is designed to equip you with critical skills in digital defense and ethical hacking, taking place at our newly established Centre of Excellence Lab.



Date: 01 October 2026

Please note the two-part schedule for the morning:

- **Assembly & Briefing:**



Time: 9:00 A.M.



Venue: Digital Lab, AIML Department

- **Main Inaugural Ceremony:**



Time: 10:00 A.M.



Venue: Sir M. V. Hall

All selected students are requested to attend the inaugural function without fail. We look forward to your enthusiastic participation. 🙏

Be present on time and make the occasion a memorable one!

In Association with: ICT Academy | Redington Foundation | CISCO

About the 21-Day Training Programme

The 21-Day Cybersecurity Training Program is a premier educational initiative hosted by the SJM Institute of Technology in collaboration with global technology leaders. Designed specifically for students of the AIML Department, this intensive course bridges the gap between academic theory and real-world cybersecurity demands.

Through the newly inaugurated **Centre of Excellence Lab**, participants will engage in immersive, hands-on learning environments. The curriculum focuses on identifying vulnerabilities, defending networks, and understanding the complex landscape of modern digital threats.

Our Esteemed Partners

ICT Academy

An initiative of the Government of India

Pioneering in faculty and student development, ICT Academy bridges the industry-academia skill gap through rigorous training frameworks and industry connections.

Redington Foundation

Corporate Social Responsibility

Dedicated to enhancing educational infrastructure and providing cutting-edge technological access to emerging engineers across the nation.

CISCO

Global Technology Leader

Providing the core curriculum, simulation tools (Cisco Packet Tracer), and industry-recognized certification pathways for network security.

SJM University

Host Institution

Fostering a culture of innovation and technical excellence through the AIML Department's dedicated Digital Lab facilities.

Leadership & Dignitaries

We are honored to have the inaugural ceremony presided over and graced by distinguished leaders from academia and industry.

Dr. K. B. Gudasi

Hon. Vice-Chancellor, SJM University
Presiding Officer of the Inaugural Ceremony.

Dr. M. S. Ganachari

Registrar, SJM University
Guest of Honor representing University
Administration.

Dr. Kumaraswamy B. G.

Principal, SJM Institute of Technology
Chief Host of the Institution.

Mr. Praveen S. Gatti

Senior Project Manager, ICT Academy
Industry Guest of Honour.

Mr. Jakaula

Senior Relationship Manager, ICT Academy
Industry Guest of Honour.

HoD & Staff Coordinators

AIML Department, SJMIT
Organizing Committee for the 21-Day Program.

"Empowering the next generation of engineers with the tools to secure our digital future is not just an educational goal; it is a national imperative."

Inaugural Day Schedule

To ensure a smooth and organized event, all selected participants must adhere to the timeline detailed below for the 1st of October, 2026.

Time	Activity	Venue & Details
08:45 A.M.	Student Reporting	Digital Lab, AIML Department. Mark attendance with coordinators.
09:00 A.M.	Initial Assembly & Briefing	Digital Lab. Distribution of ID badges and program kits.
09:40 A.M.	Procession	Students and staff proceed together to the main auditorium.
10:00 A.M.	Arrival of Dignitaries	Sir M. V. Hall. Welcome address and lighting of the lamp.
10:15 A.M.	Inaugural Address	Dr. K. B. Gudasi, Hon. Vice-Chancellor.
10:35 A.M.	Keynote Presentation	ICT Academy Representatives (Mr. Praveen S. Gatti & Mr. Jakaula).
11:15 A.M.	Vote of Thanks	HoD, AIML Department.
11:30 A.M.	High Tea	Sir M. V. Hall Foyer.
12:00 P.M.	First Session Commences	Return to Digital Lab for Day 1 technical training.

Why Cybersecurity?

As the world becomes increasingly interconnected, the reliance on digital infrastructure has never been higher. From financial transactions to critical healthcare data, every aspect of modern society operates on networks that are constantly targeted by malicious actors. In this landscape, cybersecurity is no longer an optional IT function—it is the backbone of global stability and trust.

For students in the Artificial Intelligence and Machine Learning (AIML) department, understanding cybersecurity is particularly crucial. AI models are vulnerable to data poisoning, adversarial attacks, and privacy breaches. Securing these models requires a deep foundational understanding of network architecture, cryptography, and threat intelligence.

The global deficit of trained cybersecurity professionals is estimated to be over 3.4 million. This program aims to bridge that gap by equipping SJMIT students with highly sought-after industry skills.

Program Objectives

By the conclusion of this 21-day intensive training, participants will be able to:

- Understand the core principles of data confidentiality, integrity, and availability (The CIA Triad).
- Configure and secure basic network infrastructure using Cisco technologies.
- Identify common vulnerabilities in software and web applications.
- Execute ethical hacking methodologies to test system defenses.
- Respond to simulated cyber incidents in a controlled lab environment.
- Apply artificial intelligence concepts to automate threat detection.

This hands-on approach ensures that theoretical knowledge is immediately tested through practical application in the Centre of Excellence Lab, building confidence and technical competence.

Curriculum Overview: Week 1

The first week focuses on building a robust foundation in networking and basic security concepts. All sessions take place in the Digital Lab.

Days 1-2: Foundations of Digital Security

Topics Covered:

- Introduction to Cybersecurity: History and evolution of threats.
- The CIA Triad and basic risk management concepts.
- Understanding malware: Viruses, worms, trojans, and ransomware.
- Social Engineering: Phishing, spear-phishing, and human vulnerabilities.

Days 3-5: Network Architecture & Protocols

Topics Covered:

- OSI and TCP/IP Models deeply explored.
- IP Addressing, Subnetting, and DNS operations.
- Introduction to Cisco Packet Tracer (Lab Session).
- Vulnerabilities in common protocols (HTTP, FTP, Telnet).

Days 6-7: Applied Cryptography

Topics Covered:

- Symmetric vs. Asymmetric encryption (AES, RSA).
- Hashing algorithms (SHA-256) and digital signatures.
- Public Key Infrastructure (PKI) and Certificate Authorities.
- Lab: Encrypting communications and verifying file integrity.

Curriculum Overview: Week 2

Week two transitions into offensive security concepts and defensive hardening strategies.

Days 8-10: Vulnerability Assessment

Topics Covered:

- Introduction to Nmap and network scanning techniques.
- OS fingerprinting and port scanning.
- Using vulnerability scanners (OpenVAS, Nessus basics).
- Lab: Scanning the local sandbox network for open vulnerabilities.

Days 11-12: Ethical Hacking Fundamentals

Topics Covered:

- The phases of penetration testing.
- Exploitation frameworks (Introduction to Metasploit).
- Password cracking concepts and tools.
- Ethics and legal boundaries of penetration testing.

Days 13-14: Network Defense & Hardening

Topics Covered:

- Configuring Firewalls and Intrusion Prevention Systems (IPS).
- Access Control Lists (ACLs) on Cisco routers.
- Virtual Private Networks (VPNs) and secure tunneling.
- Lab: Hardening a simulated enterprise network.

Curriculum Overview: Week 3

The final week emphasizes incident response, AI in security, and the capstone assessment.

Days 15-17: Web Application Security

Topics Covered:

- OWASP Top 10 Vulnerabilities overview.
- SQL Injection (SQLi) and Cross-Site Scripting (XSS).
- Securing APIs and microservices.
- Lab: Identifying flaws in a deliberately vulnerable web app.

Days 18-19: Incident Response & Forensics

Topics Covered:

- The Incident Response Lifecycle (Preparation, Detection, Eradication, Recovery).
- Analyzing log files and identifying anomalies using AI/ML techniques.
- Basic digital forensics and chain of custody.

Days 20-21: Capstone Project & Assessment

Topics Covered:

- Red Team vs. Blue Team simulation exercise.
- Final practical examination in the Centre of Excellence Lab.
- Course review, feedback, and certification guidelines.

Code of Conduct & Lab Guidelines

The Centre of Excellence Lab is a professional environment equipped with advanced hardware and software. To ensure a productive and safe learning experience, all participants must strictly adhere to the following guidelines throughout the 21-day program.

Mandatory Attendance Policy:

Attendance is strictly monitored. As stated in the invitation, all selected students are required to attend the inaugural function without fail, and maintain a minimum of 90% attendance throughout the 21 days to be eligible for certification.

General Rules

- **Punctuality:** Students must be seated in the Digital Lab 10 minutes before the session begins. Latecomers will not be permitted to interrupt the session.
- **Professional Decorum:** Maintain silence during lectures. Use the designated Q&A periods to ask questions.
- **Equipment Care:** Do not alter hardware configurations, disconnect cables, or install unauthorized software on lab machines.
- **Ethical Behavior:** The hacking techniques taught in this course are strictly for educational purposes. Any attempt to use these skills maliciously against the university network or external targets will result in immediate expulsion from the program and disciplinary action.
- **Data Privacy:** Do not save personal files on lab computers. Use authorized cloud storage or designated USB drives as instructed by the trainer.

Dress Code

Students must adhere to the standard SJM Institute of Technology dress code on all days. ID cards must be worn visibly at all times within the Centre of Excellence Lab.

Certification & Next Steps

Upon successful completion of the 21-Day Hands-on Training Programme, eligible students will receive a joint certification from SJM University, ICT Academy, and CISCO. This credential serves as a powerful testament to your practical skills and dedication to the field of cybersecurity.

Evaluation Criteria

Certification is not automatically granted for attendance alone. Students will be evaluated based on:

- Daily lab practical completions (30%)
- Mid-program theoretical quiz (20%)
- Final Capstone Project / Simulation (50%)

"Cybersecurity is a shared responsibility, and it boils down to this: in cybersecurity, the more systems we secure, the more secure we all are."

We are thrilled to embark on this 21-day journey with you. The skills you learn here in the Digital Lab will lay the foundation for a resilient and secure digital future.

Contact Information:

For any queries regarding the schedule or lab access, please reach out to the Staff Coordinators at the AIML Department.

Welcome to the Program!